
Cognitive Interpretability of Foundation Models

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1 Workshop Content and Topics

Tagline: Building Cognitive Interpretability as the unified science of foundation model cognition, behavior, and internal representations.

Understanding latent cognitive capabilities, behaviors, and learned internal representations of foundation models has emerged as a central challenge in AI research, with broad implications for the science of intelligence, ML theory, and AI alignment. Existing research has addressed this challenge in two complementary ways, which have largely followed independent trajectories. First, empirical and computational methods from cognitive science offer methodological tools for reverse-engineering intelligent behavior – such as psychometric theory, construct validation, and diagnostic experimental design. These methods can be adapted to diverse intelligences, including artificial systems, even if their cognitive properties diverge from humans or other animals [11]. Second, computational neuroscience and mechanistic interpretability offer tools for identifying internal model circuits and representational structure that underpin it.

Why now This workshop is timely given rapid developments in foundation model capabilities and architecture, that lag behind our ability to understand them. The methodological fragmentation across cognitive science, computational neuroscience, and mechanistic interpretability is a missed opportunity to accelerate scientific understanding of foundation models, and shape their societal impacts and in a beneficial way. Our workshop will provide a dedicated venue for these research communities to build new collaborations and establish common methodological standards under the umbrella of **cognitive interpretability**: the systematic study of foundation model behavior and representations.

Cognitive capabilities of foundation models. Foundation models have emerged as useful **model organisms** for the science of intelligence — systems that allow systematic manipulations of task conditions to isolate computational principles driving behavior. Recent work used foundation models to replicate human planning and gameplay [10, 6], automatically discover interpretable cognitive models [5], and inform new theories of cognition [14, 7]. Foundation models also serve as components in larger computational frameworks, in particular as models of human-like inductive biases in generalizable world models [10, 15, 12, 13]. Several works focus on differences between human-AI cognition, to understand how model capabilities emerge, and to diagnose the underlying representations [8, 4]. Despite this we are still lacking a principled measurement frameworks for foundation models [16] whose cognitive capabilities may fundamentally differ from those of humans, or express properties invisible to human-centric frameworks.

Understanding internal representations. Just as zebra finch models in biology provide access to neural circuits difficult to study in more complex animals, foundation models expose internal states that can be conveniently experimented with at scale. LLMs fine-tuned on behavioral data have been shown to develop internal representations aligned with neural activity [2]. Large reasoning models match human behavioral patterns in game learning and predict brain activity better than deep RL alternatives [6]. Recent studies [1] examine model’s internal representations to understand which concepts are learned by the model [9, 3], with implications to AI oversight and alignment. These tools offer a complementary lens on which internal states can be measured.

The disciplines of neuroscience, mechanistic interpretability and cognitive science share the common goal to understand intelligent systems, yet collaboration across these sub-fields remains limited. This workshop will enable them to share perspectives, develop a unified research agenda, and establish new collaborations toward solving a shared set of **open problems**, including:

1. distinguishing between retrieval and emergent computation in foundation models – a longstanding question in ML theory and the science of intelligence;

2. developing developing construct-valid cognitive evaluation frameworks for foundation models, with implications for AI oversight and alignment;
3. formally characterizing differences in human-AI cognition toward building more effective AI assistants;
4. characterizing the cognitive-theoretic basis of systematic versus brittle generalization in foundation models, with implications for predicting model behavior under distribution shift

We invite speakers and contributed papers focusing on the following themes:

- Comparing human and LLM-based agents under matched task constraints to characterize similarities and differences in behaviour and representations
- Extracting latent inductive biases and characterizing emerging cognitive capabilities in foundation models
- Using foundation models to test theories of cognition at scale
- Examining human-AI performance gaps as diagnostic signals for representational structure
- Representational similarity analysis and neural alignment as tools for model interpretation
- Using computational cognitive models as baselines for evaluating and interpreting foundation model behavior, i.e. causal reasoning or abstraction learning
- Identifying internal structures that support cognitive-like computations in foundation models

Prior Related Workshops and Points of Difference Prior workshops addressed adjacent themes, but focused on cognitive science, neuro-AI, or mechanistic interpretability in isolation. For instance, ReAlign (ICLR 2023–2025), UniReps and NeuroAI (NeurIPS 2024), and MechInterp (ICML 2024–2025) all illustrate the methodological fragmentation issue identified in our motivation. The closest prior workshop is LLMs and Cognition (ICML 2024), which focused on benchmarking LLM performance on cognitive tasks. Our workshop is organized by an independent group and reflects two years of rapid field development. Our distinct focus on developing a unified science of foundation models by integrating methodological traditions to bridge the behavior vs. representations gap, has not been addressed in prior workshops.

2 Invited Speakers

Weinan Sun (Cornell University, USA) **Confirmed**

<https://nbb.cornell.edu/weinan-sun>

Potential topics: Compositional generalization in foundation models

Marcel Binz (Helmholtz Munich, Germany) **Confirmed**

<https://marcelbinz.github.io/>

Potential topics: In search of the cognitive core

Chris Summerfield (Oxford and AISI, UK) **Confirmed**

https://csummerfield.github.io/personal_website/

Potential topics: Bridging biological and artificial reasoning: implications for AI evaluation and alignment

Kimberly Stachenfeld (Google Deepmind, USA) **Confirmed**

<https://neurokim.com/>

Potential topics: Automated cognitive modeling for humans and animals with foundation models

Tessa Verhoef (Leiden University, Netherlands) **Confirmed**

<https://www.universiteitleiden.nl/en/staffmembers/tessa-verhoef>

Potential topics: Inductive and gestalt biases in foundation models

Eric Schulz (Helmholtz Munich, Germany) **Confirmed**

<https://hcai-munich.com/eric.html>

Potential topics: Toward measurement science in AI

Mor Geva (Tel Aviv University, Israel) **Invited**

<https://mega002.github.io/>

Potential topics: Mechanistic interpretability of knowledge in language models

Panelists: Been Kim (Google Deepmind, USA), Marcelo Mattar (New York University, USA), Chris Summerfield (Oxford and AISI, UK), Mor Geva (Tel Aviv University, Israel)

3 Logistics

Concurrent NeurIPS 2026 workshop proposals Organizers are not submitting concurrent proposals

Location preference Paris preferred. (Atlanta as a second alternative)

Advisory Committee The organizers are supported by a steering committee including Pierre-Yves Oudeyer (Inria Center of University of Bordeaux), and Leyla Isik (Johns Hopkins University).

Tentative Schedule Our schedule includes 7 keynote talks, two poster sessions and 3 spotlight talks selected from outstanding paper submissions. Each keynote talk will be 30 minutes followed by a 5-min Q&A. A workshop website displaying the detailed schedule will be made upon workshop acceptance.

Table 1: A tentative schedule.

08:00 am – 08:30 am	Poster setup
08:30 am – 08:40 am	Opening remarks
08:40 am – 09:15 am	Keynote 1
09:15 am – 09:50 am	Keynote 2
09:50 am – 10:30 am	Poster session I (40 min) & Coffee break
10:30 am – 11:05 am	Keynote 3
11:05 am – 11:40 am	Keynote 4
11:40 am – 12:00 pm	Spotlight talks (3 5-min talks + 5 min Q&A)
12:00 pm – 01:30 pm	Lunch & Posters
01:30 pm – 02:05 pm	Keynote 5
02:05 pm – 02:40 pm	Keynote 6
02:40 pm – 03:30 pm	Poster session II (50 min) & Coffee break
03:30 pm – 04:05 pm	Keynote 7
04:05 pm – 05:00 pm	Panel discussion
05:00 pm – 05:10 pm	Concluding remarks

Tentative Timeline Submission: August 29, 2026, AoE; decisions: September 22, 2026, AoE; camera-ready deadline: October 8, 2026 (AoE); the workshop date: December 12 or December 13, 2026.

Attendees This new workshop regroups communities from cognitive science, mechanistic interpretability, and neuroscience, who work with foundation models. Considering their substantial growth at NeurIPS in recent years, we estimate approximately 200 attendees.

Conflicts of Interest We will ensure that organizers will not be involved in reviewing submissions from someone within the same organization.

Dissemination and Discussion We will advertise the workshop and the call for papers on social media (research community slack channels, LinkedIn, X) and mailing lists (i.e. cvnet). Talks will be recorded and posted on the workshop website. To promote discussion among participants we include time for audience questions after each talk, two poster sessions, and a panel discussion.

To initiate a sustained cross-disciplinary effort, we will form a working group during the event that will collaborate on a **post-workshop position paper** summarizing key insights and proposing a joint research agenda. This foundations for this paper will be seeded by a panel, moderated by one of the workshop organizers, and featuring four invited panelists (Summerfield, Kim, Geva, Mattar). To panel will be opened with 3 questions drawn from the workshop’s core themes, circulated to panelists in advance. Audience participation will be structured through a shared question submission form open during the morning sessions, where attendees can upvote questions throughout the workshop.

Technical Needs Provided speaker consent, the workshop presentations will be recorded and made publicly available. We require a projector, a microphone, and a setup for video recording.

4 Diversity and Inclusion

Our organizers bring together diverse institutions across the USA, France, Canada, Singapore, and the UK, and complementary expertise in ML, AI, cognitive Science, and neuroscience. The proposed speakers represent diverse perspectives from industry and academia, covering neural modeling, cognitive modeling, and model interpretability. The list of speakers is gender-balanced, and includes early-career and senior researchers from 5 different countries (USA, UK, Germany, Israel, and the Netherlands). We will broadcast workshop announcement to minority communication channels (Black in AI, Women in AI, LatinX in AI) to encourage diverse participation. We will recruit a diverse and gender-balanced group of reviewers at various stages of their careers. Program committee will cover all topics and include reviewers from academia and industry. At least 20 reviewers will be graduate students.

Organizers

- **Marta Kryven** (Assistant Professor at Dalhousie University, Canada)
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Contact: marta.kryven@dal.ca
Bio: Marta Kryven directs the Computation and Cognition lab where she studies cognitive interpretability of foundation models, including using LLM-based agents to study how people formulate hypotheses under uncertainty, make plans, and learn new concepts. Her work received two Spotlight Awards at NeurIPS 2025 (main track and LAW Workshop) for empirical work using LLMs in models of cognition, and an Annual Research Highlight in Nature Computational Science 2021 for computational cognitive science work. She is a lead co-chair at CogSci 2027 conference, has led organizing of the Canadian Computation and Cognition Symposium in 2025 and 2026, and served on the program committee of Atlantic AI Symposium (2025, 2026).
- **Yewen Pu** (Assistant Professor at Nanyang Technological University, Singapore)
<https://evanthebouncy.github.io/>
Contact: yewen.pu@ntu.edu.sg
Bio: Yewen Pu is an assistant professor at Nanyang Technological University, where he leads the Natural Programming Lab. His research focuses on code generation, instruction-following agents, and collaborative design, with broader interests in human-AI communication. He has received the Nanyang Assistant Professorship (1.5M SGD), and a PI of “Expert-in-the-Loop Agentic AI for Knowledge Discovery and Evolution in Manufacturing”, a joint project with Agency for Science, Technology and Research (A*Star). Previously, he was a Principal Research Scientist at Autodesk Research.
- **Cedric Colas** (Inria, France)
<https://cedriccolas.com/>
Contact: cedric.colas@inria.fr
Bio: Cédric Colas is a Marie Skłodowska-Curie Postdoctoral Fellow working at the intersection of artificial intelligence and cognitive science. His research focuses on autotelic agents: artificial agents that can generate their own goals, pursue them autonomously, and progressively expand their repertoires of skills and behaviors. He studies how such agents can learn through interaction with humans, their environments, and other agents, with a particular interest in open-ended learning trajectories and cumulative cultural evolution. His work integrates questions and methods from reinforcement learning, cultural evolution, program synthesis, evolutionary search, and large language models. He co-organized workshops on Intrinsically Motivated Open-ended Learning (IMOL) and Language and Reinforcement Learning, as a Program chair for the IMOL conference 2025, and served as an editor for an interdisciplinary special issue on Goal Dynamics.
- **Momchil Tomov** (Harvard University, USA)
<https://momchiltomov.com/>
Contact: mtomov@g.harvard.edu
Bio: Momchil Tomov is an incoming Assistant Professor in Psychology & Neuroscience and Computer Science (joint appointment) at Boston College. His research focuses on naturalistic learning and decision making in humans and machines. In one line of research, he investigates how the brain learns mental models of the world (referred to as *world models* in AI) in order to plan and make decisions in video games. In another line of research, he applies insights from naturalistic human decision making to improving planning algorithms for self-driving cars. He has co-organized journal clubs on decision making, large-scale hackathons, and events featuring high-profile speakers such as Jack Dorsey and Sheryl Sandberg.
- **Ivan Titov** (University of Edinburgh, UK and University of Amsterdam, Netherlands)
<http://ivan-titov.org>
Contact: ititov@inf.ed.ac.uk
Bio: Ivan Titov is a Professor of Natural Language Processing and ML at the Universities of Edinburgh and Amsterdam, where he directs the Centre for Doctoral Training in NLP and co-directs the ELLIS NLP program. His research focuses on interpretability, trustworthiness, and controllability of LLMs, supported by an ERC grant and NWO Vici fellowship. He is a Turing and ELLIS Fellow, has received outstanding paper awards at ACL and EMNLP, and served as Program Co-Chair at ICLR 2021 and CoNLL 2018.

Program Committee

We will recruit PC members with diverse backgrounds and encourage qualified authors of submissions to volunteer as reviewers. We expect 50 submissions. To ensure 3 reviews per paper, we plan to recruit 60 PC members in total. Each member will review no more than 3 submissions.

Confirmed PC members include:

- Dr. Sugandha Sharma (NVIDIA)
- Dr. Frank Rudzicz (Dalhousie)
- Dr. Hassan Sajjad (Dalhousie)
- Dr. Yen-Ling Kuo (UVA)
- Dr. Yuanfang Zhao (JHU)
- Dr. Tao Gao (UCLA)
- Dr. Max van Duijn (Leiden)
- Lance Ying (Harvard)
- Matteo Bortoletto (Stuttgart)
- Jeff Wang (JHU)
- Yifan Yin (JHU)
- Jieneng Chen (JHU)
- Mung Yao Jia (UIUC)
- Shunchi Zhang (JHU)
- Zhining Zhang (PKU)
- Kathy Garcia (JHU)
- Hannah Small (JHU)
- Manasi Malik (JHU)
- Zirui Chen (JHU)
- Theo Sechopoulos (HealthBook+)
- Alex Gu (MIT)
- Hao Tang (Cornell)
- Martin Ma (UMich)
- Claas Beger (Cornell)
- Tom Kouwenhoven (Leiden)
- Ryan Hardesty Lewis (MIT)
- Top Piriyaakulkij (Cornell)
- Sam Acquaviva (Stealth Startup)

We are inviting more PC members. Here are the tentative PC members:

- Dr. Desmond Ong (UT Austin)
- Dr. Xavier Puig (Meta)
- Dr. Lio Wong (Stanford)
- Dr. Katie Collins (MIT)
- Dr. Olga Golovneva (Meta)
- Dr. Jack Lanchantin (Meta)
- Dr. Bernhard Egger (Friedrich-Alexander-Universität Erlangen-Nürnberg)
- Dr. Kohitij Kär (York University)
- Dr. Gemma Roig (Goethe University Frankfurt)
- Dr. Kelsey Allen (UBC)
- Dr. Mark Ho (NYU)
- Dr. Ratan Murty (Georgia Tech)
- Dr. Martin Schrimpf (EPFL)
- Dr. Anna Ivanova (Georgia Tech)
- Dr. Frederick Callaway (NYU)
- Dr. Aidan Curtis (MIT)
- Dr. Tuan Anh Le (Google)
- Dr. Ilia Sucholutsky (NYU)
- Dr. Zenna Tavares (Basis AI, Columbia)
- Zhutian Yang (MIT)
- Ruocheng Wang (Stanford)
- Andi Peng (MIT, Anthropic)
- Kartik Chandra (MIT)
- Kanishk Gandhi (Stanford)
- Brian Zhang (JHU)
- Binze Li (UCLA)
- Yan Zheng (UT Austin)
- Bo Liu (NUS)
- Kai Zhang (OSU)
- Xuehai He (UCSC)
- Zhihan Liu (Northwestern)

References

- [1] Emmanuel Ameisen, Jack Lindsey, Adam Pearce, Wes Gurnee, Nicholas L. Turner, Brian Chen, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. Circuit tracing: Revealing computational graphs in language models. *Transformer Circuits Thread*, 2025. URL <https://transformer-circuits.pub/2025/attribution-graphs/methods.html>.
- [2] Marcel Binz, Elif Akata, Matthias Bethge, Franziska Brändle, Fred Callaway, Julian Coda-Forno, Peter Dayan, Can Demircan, Maria K Eckstein, Noémi Éltető, et al. Centaur: a foundation model of human cognition. *arXiv preprint arXiv:2410.20268*, 2024.
- [3] Thomas Bush, Stephen Chung, Usman Anwar, Adrià Garriga-Alonso, and David Krueger. Interpreting emergent planning in model-free reinforcement learning. *arXiv preprint arXiv:2504.01871*, 2025.
- [4] Leonardo Hernandez Cano, Ivan Zareski, Luisa El Amouri, Pinzhe Zhao, Max Mascini, Emanuele Sansone, Yewen Pu, Bonan Zhao, and Marta Kryven. Prospective compression in human abstraction learning. *arXiv preprint arXiv:2605.09985*, 2026.
- [5] Pablo Samuel Castro, Nenad Tomasev, Ankit Anand, Navodita Sharma, Rishika Mohanta, Aparna Dev, Kuba Perlin, Siddhant Jain, Kyle Levin, Noémi Éltető, et al. Discovering symbolic cognitive models from human and animal behavior. *bioRxiv*, pages 2025–02, 2025.
- [6] Botos Csaba, Sreejan Kumar, Austin Tudor David Andrews, Laurence Hunt, Chris Summerfield, Joshua B Tenenbaum, Rui Ponte Costa, Marcelo G Mattar, and Momchil Tomov. Reason to play: Behavioral and brain alignment between frontier lrms and human game learners. *arXiv preprint arXiv:2605.08019*, 2026.
- [7] Michael C Frank and Noah D Goodman. Cognitive modeling using artificial intelligence. 2025.

- [8] Kathy Garcia, Emalie McMahon, Colin Conwell, Michael Bonner, and Leyla Isik. Modeling dynamic social vision highlights gaps between deep learning and humans. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=wAXsx2MYgV>.
- [9] Wes Gurnee and Max Tegmark. Language models represent space and time. *arXiv preprint arXiv:2310.02207*, 2023.
- [10] Marta Kryven, Cole Wyeth, Aidan Curtis, and Kevin Ellis. Cognitive maps are generative programs. *Cognitive Sciences Society*, 2025.
- [11] Michael Levin. The computational boundary of a “self”: developmental bioelectricity drives multicellularity and scale-free cognition. *Frontiers in psychology*, 10:493866, 2019.
- [12] Top Piriyaakulkij, Yichao Liang, Hao Tang, Adrian Weller, Marta Kryven, and Kevin Ellis. Poe-world: Compositional world modeling with products of programmatic experts. *Advances in Neural Information Processing Systems*, 38:26609–26638, 2026.
- [13] Wasu T Piriyaakulkij, Cassidy Langenfeld, Tuan A Le, and Kevin Ellis. Doing experiments and revising rules with natural language and probabilistic reasoning. *Advances in Neural Information Processing Systems*, 37:53102–53137, 2024.
- [14] Milena Rmus, Akshay K Jagadish, Marvin Mathony, Tobias Ludwig, and Eric Schulz. Towards automation of cognitive modeling using large language models. *arXiv preprint arXiv:2502.00879*, 2025.
- [15] Hao Tang, Darren Key, and Kevin Ellis. Worldcoder, a model-based llm agent: Building world models by writing code and interacting with the environment. *Advances in Neural Information Processing Systems*, 37:70148–70212, 2024.
- [16] Konstantinos Voudouris, Mirko Thalmann, Alex Kipnis, José Hernández-Orallo, and Eric Schulz. Measuring what ai systems might do: Towards a measurement science in ai. *arXiv preprint arXiv:2603.00063*, 2026.